

Day 2: Sys admin

Answers

Log 1 — the server

What is going on?

Log files were never cleared since January. They grew until the disk was full (71% to 100% in three days). The disk filled, the backup failed, the website stopped and today's data was not saved.

How would you resolve it?

Delete or archive the old log files to free space, restart the website, then run a backup and check it worked.

What would stop it happening again?

Turn on log rotation so logs clear themselves, and alert someone when the disk passes 85%.

Log 2 — the backups

What is going on?

The off-site copy failed for 8 days and nobody noticed. The backup drive was also showing read errors, then it died — so every copy was on one broken disk.

How would you resolve it?

Replace the drive and restore from the off-site copy, then check the restored files actually open.

What would stop it happening again?

Alert on the first failed off-site copy, keep two copies in two places, and test a restore regularly.

Log 3 — the website

What is going on?

The security certificate expired. It warned for 30 days, but the reminders went to an unread shared mailbox and the webmaster was on leave with no cover. Visitors get a security warning, the app won't sync, and the tide gauges can't upload — SAM-02 starts overwriting old readings.

How would you resolve it?

Renew and install the certificate. The real hold-up is the purchase approval, so escalate to someone who can approve it, and copy SAM-02's readings off before they are lost.

What would stop it happening again?

Set renewal to happen automatically, and put expiry dates on the maintenance plan with a named person — plus leave cover, including approvers.

Log 4 — the database

What is going on?

Rows were deleted from obs_daily by mistake, then the wrong backup — the staff directory's — was restored into the clide database. CliDE now has the wrong tables and won't start: obs_daily and obs_aws do not exist. The clue: the restore took 5 minutes, not 90.

How would you resolve it?

Restore the correct file, clide_2026-06-17.dump (~90 minutes), then re-enter the 8 hours of observations no backup holds.

What would stop it happening again?

Clear names and separate folders per application, a backup taken before any restore, and a tested restore each quarter.

Maintenance schedule**Daily**

Is the website up? Did last night's backup run? Any red lines in the logs? — Duty officer

Weekly

Disk space and log file sizes; did the off-site copy arrive?; install security updates — IT officer

Monthly

Certificate and domain expiry (renew anything under 45 days); remove leavers' accounts — IT officer

Quarterly

Restore a backup onto a test server and confirm the app starts; re-read and correct the plan — IT manager

Annually

Full recovery drill; review licences and support contracts; replace ageing disks — IT manager and supervisor